MAGIC: Multi-Grained Conditional Diffusion for High-Fidelity PPG-to-ECG Translation

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***Abstract*— Building a cardiac digital twin relies on high fidelity electrocardiogram signals to accurately simulate individual cardiovascular function. However, conventional ECG acquisition is not well suited for long term, unobtrusive home monitoring, while photoplethysmography, despite its advantage in continuous collection, lacks electrophysiological information. To bridge these complementary modalities for computational analysis, we propose MAGIC, a multi-grained conditional diffusion framework for generating ECG-like waveforms from PPG. MAGIC separates conditional information into a global condition and token-wise local conditions: the former summarizes segment-level cardiovascular context and modulates the diffusion Transformer through adaptive layer normalization, while the latter preserves patch-level PPG cues and is injected through gated cross-attention. The conditional encoder is pre-trained with paired PPG and ECG signals using ROI-weighted reconstruction, contrastive learning, and latent alignment, so that PPG-derived conditions are explicitly encouraged to match ECG-derived representations. Across four public datasets and four downstream proxy tasks, MAGIC improves distributional fidelity in most settings and shows competitive task-level utility. In representative analyses, pre-training reduces FD from 0.908 to 0.693 on MIMIC-AFib; MAGIC also achieves lower FD with 50 sampling steps than RDDM with 500 steps.**

***Index Terms*— Cardiac digital twin, Cross-modal generation, Diffusion model.**

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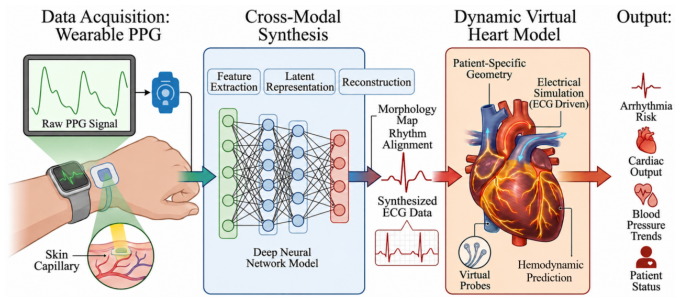

Fig. 1. A cardiac digital twin framework that leverages wearable PPG signals to generate ECG-like auxiliary signals for computational analysis.

I. Introduction

ELECTROCARDIOGRAPH (ECG) serves as a foundational data source for constructing dynamic virtual cardiac models that enable simulation and prediction of individual cardiovascular function [1]–[3]. Conventional ECG acquisition, however, relies on trained personnel and specialized equipment, limiting its applicability for continuous monitoring. Even wearable ECG devices typically require active user engagement (e.g., finger contact) and are not well suited for long-term, passive home monitoring [4], [5]. In contrast, photoplethysmography (PPG) leverages optical sensors embedded in widely available wearable devices to enable noninvasive, continuous acquisition of peripheral blood flow signals, offering substantial advantages in terms of accessibility [6], [7]. Nevertheless, PPG reflects only hemodynamic changes and lacks direct electrophysiological information, which restricts its diagnostic specificity for many cardiovascular conditions.

Faced with the trade-off between wearable convenience and ECG-specific electrical information, cross-modal translation from PPG to ECG provides a promising auxiliary route for computational cardiovascular monitoring, as shown in Figure 1. The goal is not to replace clinically acquired ECG, but to generate ECG-like signals that preserve task-relevant rhythm and morphology cues when only PPG is available. Early efforts centered on signal processing and dictionary learning [15]. These methods offered interpretability and computational efficiency but struggled to capture the complex nonlinear relationships between the two modalities. With the rise of deep learning, neural generative models have become mainstream. Recurrent and convolutional networks [18] are concise but limited in capturing long-range dependencies. Generative adversarial networks [20] suffer from training instability and mode collapse. Diffusion models [26] excel at high-fidelity synthesis. RDDM, for instance, balances global structure and local details via ROI-guided forward processes and a decoupled reverse design.

However, existing diffusion models still exhibit two limitations when applied to PPG-to-ECG translation. **(i) Insufficient cross-modal alignment in conditional guidance.** Current methods often inject PPG conditions after generic feature extraction, without explicitly encouraging PPG-derived representations to match the paired ECG representation at either the segment or patch level. This weakens the link between the conditioning signal and the target waveform. **(ii) Insufficient separation between global and local control.** A single condition pathway may mix segment-level rhythm context with patch-level morphology cues, making it unclear how global cardiovascular state and local temporal patterns respectively influence denoising. Such ambiguity can degrade QRS sharpness, rhythm timing, and distributional fidelity in the generated ECG-like waveforms.

To address these challenges, we propose MAGIC, a multi-grained conditional diffusion model. First, we design a Transformer-based conditional encoder that produces two representations from PPG: a global embedding obtained from four learnable global tokens and token-wise embeddings retained from local PPG patches. During pre-training, an auxiliary ECG encoder with the same architecture acts as a modality teacher; ROI-weighted reconstruction, contrastive learning, and latent alignment jointly encourage the PPG encoder to produce ECG-aligned global and token-wise conditions. Second, we develop a diffusion Transformer decoder in which the global condition modulates self-attention and MLP pathways via AdaLN, whereas token-wise PPG conditions are injected by gated cross-attention. This design makes the role of each conditional granularity explicit. Third, we use a velocity-prediction objective and classifier-free guidance as a practical trade-off between stability, distributional fidelity, and sampling efficiency. To evaluate the generated ECG-like waveforms, we assess generation metrics and four downstream proxy tasks: heart rate estimation, atrial fibrillation detection, diabetes detection, and blood pressure estimation.

The main contributions are summarized as follows:

- We propose a multi-grained conditional encoder that separates segment-level global context from patch-level local PPG information. Four learnable global tokens summarize the PPG segment, while the remaining patch tokens retain local temporal cues. A paired ECG encoder, ROI-weighted reconstruction, contrastive learning, and latent alignment are used to pre-train ECG-aligned PPG conditions.
- We introduce an adaptive condition-injection strategy for a 1D diffusion Transformer. The global condition enters the model through AdaLN together with the timestep embedding, whereas token-wise conditions enter through a learned gated cross-attention module, clarifying how global and local conditions contribute differently to denoising.
- We provide a balanced evaluation of MAGIC across generation metrics, inference time, and downstream proxy tasks. The results show clear advantages in FD in most benchmark settings and improved sampling efficiency.

II. Related Work

Early work relied on signal processing and dictionary learning. Zhu et al. [15] proposed a linear DCT-domain model to map PPG to ECG via regression, which worked well for subject-specific models but generalized poorly due to linear assumptions. Tian et al. [16] introduced cross-domain joint dictionary learning (XDJDL), jointly optimizing sparse dictionaries and their linear mapping, with a label-consistent extension (LC-XDJDL) incorporating disease labels for improved discriminative power. While interpretable and efficient, these methods struggle to capture complex nonlinear relationships between PPG and ECG.

Deep neural network-based generative methods have become mainstream and can be categorized into three classes: **(i) Basic recurrent/convolutional methods** leverage temporal modeling for end-to-end generation. Tang et al. [17] used bidirectional LSTM for sequence-to-sequence mapping. Murmu et al. [18] proposed a memory-augmented autoencoder with MLPNN feature translation and LSTM-RNN synthesis. These models are structurally simple but have limited capacity for long-range dependencies, resulting in suboptimal fidelity. **(ii) Adversarial methods** employ GANs for more realistic waveform generation. Sarkar et al. [19] introduced CardioGAN for unpaired translation with attention and dual discriminators. Vo et al. [20] used conditional Wasserstein GAN to improve training stability. Skoric et al. [21] extended GANs to multi-modal generation from a single sensor. Despite producing realistic outputs, GAN-based approaches remain prone to training instability and mode collapse. **(iii) Advanced generative methods based on Transformers and diffusion models** represent the latest progress [39]. Lan [22] proposed Performer with a shifted patch attention mechanism for multi-scale feature extraction via hierarchical structures. Yuan et al. [23] further introduced a cycle-aware module that reformulates 1D signals into 2D matrices to construct multi-semantic views of cardiac cycles. Diffusion models have gained attention for their superior quality. Adib et al. [24] first applied an improved DDPM to unconditional ECG generation. Xia et al. [25] employed diffusion models for PPG denoising, effectively handling non-Gaussian noise. Shome et al. [26] proposed RDDM with ROI-guided noise and decoupled reverse process. Belhasin et al. [27] addressed the ill-posed nature of the problem by generating multiple plausible ECG samples and averaging their classification scores to approximate the optimal classifier. However, many existing diffusion models do not explicitly separate global rhythm-level conditioning from local patch-level conditioning, nor do they clearly align PPG-derived conditions with paired ECG representations before diffusion training. MAGIC addresses this gap by combining cross-modal conditional pre-training with separate global AdaLN modulation and gated token-wise cross-attention.

III. Multi-Grained Conditional Diffusion Model

Denoising diffusion probabilistic models (DDPMs) provide a principled foundation for conditional signal generation. To

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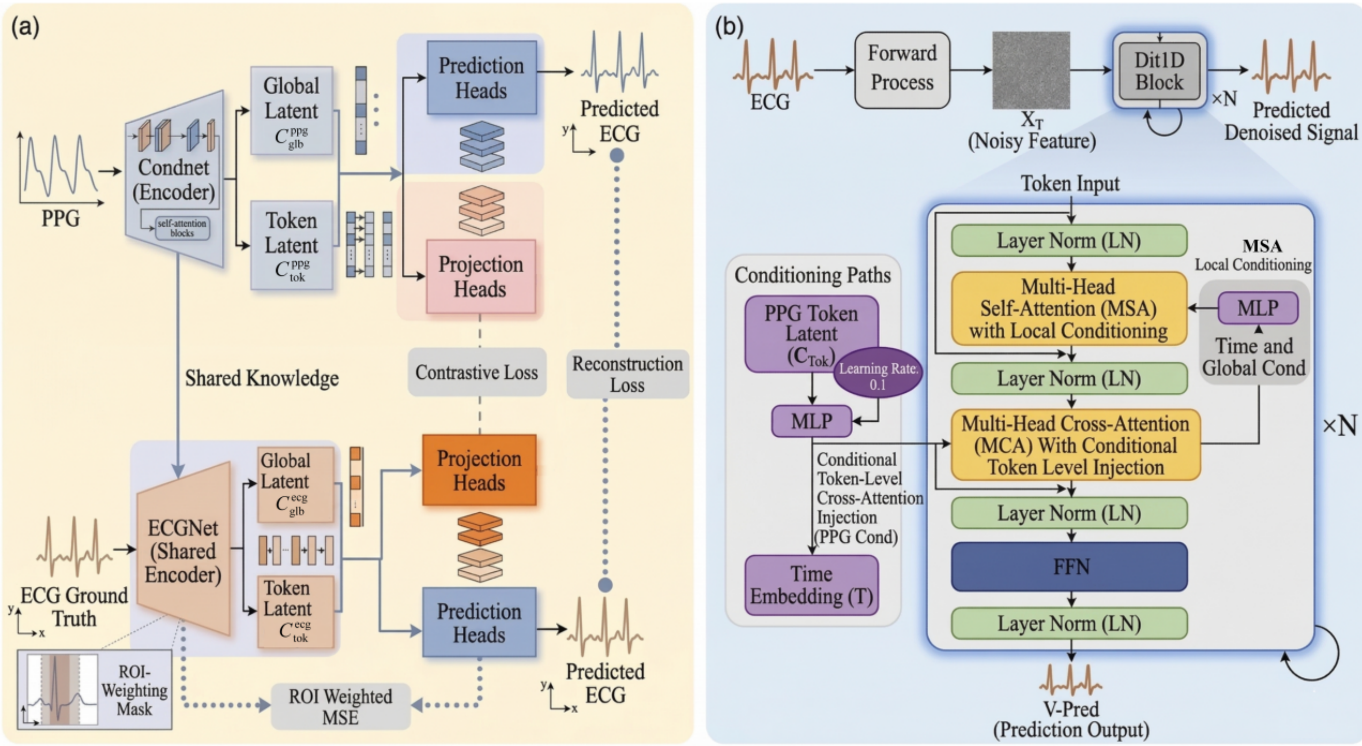


Fig. 2. Overview of MAGIC. (a) The conditional encoder extracts four-token global context and token-wise local PPG conditions, and is pre-trained using paired PPG-ECG alignment. (b) The diffusion Transformer uses global AdaLN modulation and gated cross-attention to inject local PPG conditions under the v-prediction objective.

enable PPG-guided ECG-like waveform generation, we extend this paradigm with a multi-grained conditional encoder that captures both segment-level context and patch-level temporal features from the PPG waveform. The overall architecture is illustrated in Figure 2, consisting of a conditional encoder that produces global and token-wise embeddings and a 1D diffusion Transformer (DiT) that progressively denoises the ECG signal under these two distinct conditional pathways.

A. Preliminaries: Diffusion Models

Diffusion models [8] define a forward Markov chain that gradually adds Gaussian noise to a clean data sample $x_0 \sim q(x_0)$ over $\mathcal{T}$ timesteps. For a given variance schedule $\beta_1, \cdots, \beta_{\mathcal{T}}$, the forward process is:

$$q\left(x_t \mid x_{t-1}\right) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t \mathbf{I}\right), \quad (1)$$

$$q\left(x_{\mathcal{T}} \mid x_0\right) = \prod_{t=1}^{\mathcal{T}} q\left(x_t \mid x_{t-1}\right), \quad (2)$$

Using the notation $\alpha_t = 1-\beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, the noisy sample at an arbitrary timestep t can be directly sampled from x_0:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (3)$$

The reverse process is a Markov chain parameterized by a neural network ϵ_θ that learns to reverse the noising. Starting from pure noise $x_{\mathcal{T}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the model iteratively denoises:

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 \mathbf{I}\right), \quad (4)$$

Training is typically performed by optimizing a simplified variant of the variational lower bound [9]. For conditional generation, we condition the model on an auxiliary signal c. The objective becomes:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \epsilon}\left[\left\| \epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, c, t\right)\right\|^2\right]. \quad (5)$$

In practice, many works adopt a v-prediction objective [10] which predicts $v = \sqrt{\bar{\alpha}_t}\epsilon - \sqrt{1-\bar{\alpha}_t}x_0$, often leading to more stable training.

B. Multi-Granularity Conditional Encoder

A key challenge in PPG-to-ECG translation is that the input PPG signal provides limited morphological information compared to the target ECG. To guide the diffusion process effectively, we propose a conditional encoder that extracts both global context and local fine-grained features from the PPG waveform. The encoder, denoted as ε_ψ, transforms the raw PPG signal c (of length L) into two complementary representations:

- **Global embedding** $c_{\text{glb}} \in \mathbb{R}^D$ captures segment-level cardiovascular context, such as rhythm tendency, amplitude envelope, and slow autonomic or hemodynamic state. This condition is used for global modulation of the diffusion network.
- **Token-wise embeddings** $c_{\text{tok}} \in \mathbb{R}^{N \times D}$ (where $N = L/p$ is the number of non-overlapping patches of size p)

retain local PPG patterns that can guide beat timing and morphology-related details during denoising. These embeddings are used as keys and values in cross-attention.

The architecture of $\mathcal{E}_\psi$ is a Transformer encoder [11] operating on patch embeddings of the PPG. Specifically, a 1D convolution with kernel size p and stride p converts the input waveform into patch tokens $\mathbf{z} = [z_1, \ldots, z_N] \in \mathbb{R}^{N\times D}$. Rather than relying on a single global summary token, MAGIC appends $K = 4$ learnable global tokens $\{g_k\}_{k=1}^{K}$ to the patch sequence. The whole sequence is encoded with Transformer layers using RoPE. Let $\mathbf{H} = \mathrm{Transformer}([z_1, \ldots, z_N, g_1, \ldots, g_K])$. The token-wise condition is obtained from the first N outputs, $c_{\text{tok}} = \mathbf{H}_{1:N}$, while the four global-token outputs are concatenated and projected to form $c_{\text{glb}} = W_{\text{glb}}[\mathbf{H}_{N+1}; \ldots; \mathbf{H}_{N+K}]$. Using multiple global tokens gives the encoder several learnable summary queries rather than forcing rhythm, amplitude, and morphology context into a single token. The encoder is trained with the cross-modal objectives described in Sec. III-D.

To enable classifier-free guidance [12] during sampling, we apply dropout on the condition during training: with probability p_{uncond}, we replace the condition with a learnable "unconditioned" token $c_\varnothing$. This allows the model to learn both conditional and unconditional denoising.

C. Diffusion Backbone with Adaptive Condition Injection

We design the denoising network ϵ_θ (or v_θ depending on the prediction type) as a 1D adaptation of the Diffusion Transformer (DiT) [13]. The input noisy ECG x_t (length L) is first divided into patches of size p and linearly projected to dimension D, resulting in a sequence of N tokens. Sinusoidal positional embeddings are added, and the sequence is processed by a series of DiT blocks. Each DiT block consists of:

- LayerNorm, followed by multi-head self-attention with rotary positional embeddings (RoPE) [14] to model dependencies among noisy ECG tokens.
- Adaptive layer normalization (AdaLN), where the timestep embedding and the global condition c_{glb} jointly generate shift, scale, and residual gate parameters for the self-attention and MLP paths. This is the global conditional pathway.
- Gated cross-attention with the token-wise condition c_{tok}. The noisy ECG tokens serve as queries, while local PPG tokens serve as keys and values. A learned gate controls the magnitude of this local update at each token.
- A feed-forward network (MLP) modulated by the same global AdaLN pathway.

Formally, for a block with input hidden states $\mathcal{H}$, the computation is:

$$\begin{aligned}
\mathcal{H}' &= \mathcal{H} + g_{\text{sa}} \odot \mathrm{SelfAttn}(\mathrm{AdaLN}(\mathcal{H}; \gamma_{\text{sa}}, \beta_{\text{sa}})),\\
\mathcal{U} &= \mathrm{CrossAttn}(\mathrm{LN}(\mathcal{H}'), c_{\text{tok}}, c_{\text{tok}}),\\
\mathcal{H}'' &= \mathcal{H}' + g_{\text{tok}}(c_{\text{tok}}) \odot \mathcal{U},\\
\mathcal{H}_{\text{out}} &= \mathcal{H}'' + g_{\text{mlp}} \odot \mathrm{MLP}(\mathrm{AdaLN}(\mathcal{H}''; \gamma_{\text{mlp}}, \beta_{\text{mlp}})).
\end{aligned} \tag{6}$$

Here $\gamma_{\text{sa}}, \beta_{\text{sa}}, g_{\text{sa}}, \gamma_{\text{mlp}}, \beta_{\text{mlp}}$, and g_{mlp} are generated from c_{glb} and t_{emb} via a small MLP, so the global condition controls the overall denoising dynamics. In contrast, c_{tok} provides local PPG keys and values for cross-attention. The learned token gate $g_{\text{tok}}(c_{\text{tok}})$ regulates how strongly local PPG evidence updates each noisy ECG token, making the local pathway controlled rather than a simple stack of standard cross-attention layers.

After all blocks, a final AdaLN layer and a linear projection map the sequence back to the original signal space, producing the prediction (either v_t, ϵ_t, or x_0 depending on the training objective). The overall diffusion model is denoted as $\mathcal{G}_\phi$.

D. Training Objectives

Training proceeds in two stages: first a pre-training stage for the conditional encoder, followed by the diffusion training stage.

Pre-training the Conditional Encoder. To ensure that the PPG encoder produces ECG-aligned representations for downstream diffusion, we jointly train it with an auxiliary ECG encoder on paired PPG-ECG data. The two encoders share the same architecture but have independent parameters: $\mathcal{E}_\psi$ processes PPG, while $\mathcal{E}_{\text{ecg}}$ processes the paired ground-truth ECG and provides a representation-level target. The pre-training objective contains ROI-weighted reconstruction, contrastive alignment, and latent alignment.

ROI-weighted Reconstruction Loss: We add lightweight prediction heads on top of the PPG encoder and the auxiliary ECG encoder. The PPG reconstruction head attempts to reconstruct the paired ECG from PPG-derived global and token-wise embeddings, forcing the PPG encoder to retain information useful for ECG-like waveform synthesis. The ECG reconstruction head regularizes the ECG teacher representation. The reconstruction loss is a weighted mean squared error that emphasizes QRS-centered regions. Given a binary ROI mask m indicating a 32-sample neighborhood around each detected R peak, we compute:

$$\mathcal{L}_{\text{recon}} = \mathbb{E}\left[\begin{aligned}&\lambda\|m \odot (\hat{x}_{\text{ecg}} - x_{\text{ecg}})\|^2\\ &+ \|(1-m) \odot (\hat{x}_{\text{ecg}} - x_{\text{ecg}})\|^2\end{aligned}\right], \tag{7}$$

where $\odot$ denotes element-wise multiplication, and $\lambda > 1$ increases the reconstruction penalty on the ROI. The ROI mask is used exclusively in this auxiliary reconstruction loss during conditional-encoder pre-training; it is not used in the DDPM forward noising process, reverse denoising update, sampling variance, diffusion training loss, or inference procedure.

Contrastive and Latent Alignment Losses: To align the PPG-derived embeddings with the corresponding ECG embeddings, we project the global and token-wise outputs of both encoders into a shared space. The contrastive loss pulls together PPG and ECG representations from the same sample while pushing apart representations from different samples. For a batch of size B, the contrastive loss is:

$$\mathcal{L}_{\text{cont}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(\mathrm{sim}(c^i_{\text{ppg}}, c^i_{\text{ecg}})/\tau)}{\sum_{j=1}^{B}\exp(\mathrm{sim}(c^i_{\text{ppg}}, c^j_{\text{ecg}})/\tau)}, \tag{8}$$

where $\mathrm{sim}(\cdot,\cdot)$ is cosine similarity, and τ is a temperature parameter. In addition, we directly align the latent global and token-wise embeddings of the PPG encoder with those of the ECG encoder using a stop-gradient on the ECG branch:

$$\mathcal{L}_{\text{align}} = \left\| c_{\text{glb}}^{\text{ppg}} - \mathrm{sg}(c_{\text{glb}}^{\text{ecg}}) \right\|_2^2 + \left\| c_{\text{tok}}^{\text{ppg}} - \mathrm{sg}(c_{\text{tok}}^{\text{ecg}}) \right\|_2^2, \quad (9)$$

where $\mathrm{sg}(\cdot)$ denotes stop-gradient. The total pre-training objective is therefore

$$\mathcal{L}_{\text{pre}} = \mathcal{L}_{\text{recon}} + \alpha\mathcal{L}_{\text{cont}} + \eta\mathcal{L}_{\text{align}}, \quad (10)$$

where α and η balance the contrastive and latent-alignment terms. After pre-training, the auxiliary ECG encoder and reconstruction/projection heads are discarded, and the PPG encoder initializes $\mathcal{E}_\psi$ for diffusion training.

Diffusion Training. We train the diffusion model $\mathcal{G}_\phi$ and the conditional encoder $\mathcal{E}_\psi$ jointly using a standard diffusion objective. Following recent work [10], we adopt the v-prediction formulation because it provides a stable overall trade-off in our experiments, although it is not the best choice for every individual FD/RMSE entry. The model $\mathcal{G}_\phi$ takes as input the noisy signal x_t, the timestep t, and the condition embeddings $(c_{\text{glb}}, c_{\text{tok}})$ from $\mathcal{E}_\psi$, and predicts v_t. The loss is:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t,x_0,c,\epsilon}\left[\|v_t - \mathcal{G}_\phi(x_t, t, \mathcal{E}_\psi(c))\|^2\right], \quad (11)$$

where $v_t = \sqrt{\bar{\alpha}_t}\epsilon - \sqrt{1-\bar{\alpha}_t}x_0$. During training, we also apply dropout on the condition as described in III-B to enable classifier-free guidance at inference. The overall training procedure is summarized in **Algorithm 1**.

Algorithm 1 Training of MAGIC

Require: Paired PPG-ECG dataset $\mathcal{D} = \{(c^i, x_0^i, m^i)\}$, diffusion timesteps $\mathcal{T}$, variance schedule $\{\beta_t\}$, pre-training epochs E_{pre}, diffusion epochs E_{diff}
Ensure: Trained conditional encoder $\mathcal{E}_\psi$ and diffusion model $\mathcal{G}_\phi$
Stage 1: Pre-train conditional encoder
Initialize $\mathcal{E}_\psi$ and $\mathcal{E}_{\text{ecg}}$ randomly, auxiliary heads $\mathcal{H}_{\text{glb}}, \mathcal{H}_{\text{tok}}$
for $e = 1$ to E_{pre} **do**
  Sample batch $\{(c, x_0, m)\}$ from $\mathcal{D}$; m is used only in Stage 1
  Compute $c_{\text{glb}}, c_{\text{tok}} = \mathcal{E}_\psi(c)$
  Compute $c_{\text{ecg, glb}}, c_{\text{ecg, tok}} = \mathcal{E}_{\text{ecg}}(x_0)$
  Reconstruct $\hat{x}_{\text{ecg}}$ from $(c_{\text{glb}}, c_{\text{tok}})$ using $\mathcal{H}_{\text{glb}}, \mathcal{H}_{\text{tok}}$
  Compute $\mathcal{L}_{\text{recon}}$ via (7), $\mathcal{L}_{\text{cont}}$ via (8), and $\mathcal{L}_{\text{align}}$ via (9)
  Update ψ and $\mathcal{E}_{\text{ecg}}$ to minimize $\mathcal{L}_{\text{pre}}$ in (10)
end for
Stage 2: Train diffusion model
Initialize $\mathcal{G}_\phi$ randomly, load pre-trained $\mathcal{E}_\psi$
for $e = 1$ to E_{diff} **do**
  Sample batch $\{(c, x_0)\}$ from $\mathcal{D}$; the ROI mask is not used for diffusion training
  Sample $t \sim \text{Uniform}(\{1, \ldots, \mathcal{T}\})$ and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$
  Compute $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$
  Compute $c_{\text{glb}}, c_{\text{tok}} = \mathcal{E}_\psi(c)$ with dropout probability p_{uncond}
  Compute prediction $v_t^{\text{pred}} = \mathcal{G}_\phi(x_t, t, c_{\text{glb}}, c_{\text{tok}})$
  Compute target $v_t = \sqrt{\bar{\alpha}_t}\epsilon - \sqrt{1-\bar{\alpha}_t}x_0$
  Update ϕ, ψ to minimize $\|v_t^{\text{pred}} - v_t\|^2$
end for

E. Inference via Iterative Denoising

At test time, given a PPG signal c, we first compute its conditional embeddings $(c_{\text{glb}}, c_{\text{tok}})$ using the trained encoder $\mathcal{E}_\psi$. To enable classifier-free guidance, we also compute the unconditional embeddings by passing c through the encoder with the dropout mechanism (or directly using the learned null token). The final guided prediction at each step is obtained as:

$$\tilde{v}_t = v_\theta(x_t, t, c_\varnothing) + w\big(v_\theta(x_t, t, c) - v_\theta(x_t, t, c_\varnothing)\big), \quad (12)$$

where w is the guidance scale. Starting from pure Gaussian noise $x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, we iteratively apply the reverse process for $t = T, \ldots, 1$ using the following update (for v-prediction):

$$\hat{x}_0 = \sqrt{\bar{\alpha}_t}x_t - \sqrt{1-\bar{\alpha}_t}\,\tilde{v}_t, \quad (13)$$

$$x_{t-1} = \frac{\sqrt{\bar{\alpha}_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\hat{x}_0 + \sigma_t\mathbf{z}, \quad (14)$$

where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for $t > 1$ and $\mathbf{z} = \mathbf{0}$ for $t = 1$. σ_t is the posterior variance, which can be set to β_t or learned. The ROI mask is not involved in this sampling update or in the sampling variance. The final output x_0 is a generated ECG-like signal conditioned on PPG. The full sampling procedure is given in **Algorithm 2**.

IV. Experiment and Performance

This section details the implementation specifics, including datasets, pre-processing, network architecture, training setup, and baseline models. We then report benchmark generation metrics and four downstream proxy evaluations. These downstream tasks assess whether generated ECG-like signals preserve task-relevant information, but they do not establish clinical diagnostic equivalence to real ECG.

A. Implementation Details

1) Datasets: To ensure robust model generalization across heterogeneous populations, we assemble a multi-corpus dataset from four publicly available sources: WESAD [28], DALIA [29], BIDMC [30], and MIMIC-AFib [31]. These datasets collectively provide diverse distributions in terms of subject activity, age, and clinical status, resulting in a total of 118 participants with near-equal gender representation.

WESAD contains ECG and PPG signals from 15 participants (aged 27.5 ± SD) during a structured protocol involving cognitive load (arithmetic tasks) and affective elicitation (video clips). Recordings exceeded one hour per subject, with ECG sampled at 700 Hz and PPG at 64 Hz.

DALIA consists of 15 participants (aged 30.6 ± SD) performing various daily activities over two-hour sessions.

Algorithm 2 Sampling from the trained model

Require: PPG signal c, guidance scale w, number of steps $\mathcal{T}$
Ensure: Generated ECG-like signal x_0

1: Compute conditional embeddings $(c_{\text{glb}}, c_{\text{tok}}) = \mathcal{E}_\psi(c)$
2: Compute unconditional embeddings $(c_{\varnothing,\text{glb}}, c_{\varnothing,\text{tok}})$ via null token
3: Sample $x_\mathcal{T} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
4: **for** $t = \mathcal{T}$ down to 1 **do**
5: Predict $v_t^{\text{cond}} = \mathcal{G}_\phi(x_t, t, c_{\text{glb}}, c_{\text{tok}})$
6: Predict $v_t^{\text{uncond}} = \mathcal{G}_\phi(x_t, t, c_{\varnothing,\text{glb}}, c_{\varnothing,\text{tok}})$
7: Apply guidance: $\tilde{v}_t = v_t^{\text{uncond}} + w(v_t^{\text{cond}} - v_t^{\text{uncond}})$
8: Estimate $\hat{x}_0 = \sqrt{\bar{\alpha}_t} x_t - \sqrt{1 - \bar{\alpha}_t}\tilde{v}_t$
9: Compute x_{t-1} using (14)
10: **end for**
11: **return** x_0

ECG and PPG signals were recorded at 700 Hz and 64 Hz, respectively.

BIDMC provides physiological recordings from 53 critically ill adults (aged 64.8 ± SD) admitted to the intensive care unit. Each session spans eight minutes, with PPG and tri-lead ECG (II, V, AVR) synchronously sampled at 125 Hz. This study focuses exclusively on lead II for ECG analysis.

MIMIC-AFib is derived from the MIMIC-III Waveform Database and provides binary annotations for atrial fibrillation. It comprises recordings from 35 critically ill adults, including 19 subjects diagnosed with AFib. Both ECG and PPG signals were acquired at a sampling rate of 125 Hz.

2) Data pre-processing: The dataset was constructed by aggregating four public repositories and segmenting the raw ECG and PPG signals into 4-second non-overlapping windows (128 Hz, 512 samples per window). Each window was individually cleaned using NeuroKit2 routines: PPG signals were filtered via `ppg_clean`, while ECG signals were processed with `ecg_clean` using the Pan–Tompkins method to reduce artifacts and baseline wander. R-peaks were detected on the cleaned ECG, and a binary region-of-interest (ROI) mask was created by setting a 32-sample neighborhood around each detected peak to 1. This mask is used only for the ROI-weighted reconstruction loss during conditional-encoder pre-training; it does not modify the diffusion forward process, reverse denoising process, sampling variance, or inference procedure. Finally, all ECG and PPG windows were normalized to $[-1, 1]$ via row-wise min-max scaling, yielding paired inputs for MAGIC training.

3) Network architecture: The architecture of MAGIC comprises two main components: a conditional encoder and a diffusion backbone. The conditional encoder processes the PPG waveform via patch embedding and a 6-layer Transformer with RoPE. It appends four learnable global tokens to the patch sequence; after encoding, these four tokens are concatenated and projected into a 512-D global condition vector, while the remaining patch outputs serve as token-wise local conditions. The diffusion backbone is a 12-layer 1D Diffusion Transformer (DiT1D) operating on ECG patches of size 32. In each DiT block, the global condition and sinusoidal time embedding modulate self-attention and MLP paths through AdaLN, whereas token-wise PPG conditions are injected by gated cross-attention.

4) Training setup: We train MAGIC on 2 × GeForce RTX 4090. The training configuration is detailed as follows. The diffusion process employs a cosine noise schedule with 500 timesteps, balancing generation quality and sampling efficiency. A batch size of 512 is used to leverage GPU parallelism while maintaining stable gradient updates. Two distinct learning rates are adopted: 2×10^{-4} for the diffusion backbone and 1×10^{-4} during the pre-training stage of the conditional encoder, both optimised with AdamW and a cosine warm-up scheduler. An exponential moving average (EMA) decay of 0.99 is applied to the diffusion model to improve sample fidelity during inference. For classifier-free guidance, a dropout probability of 0.1 is used during training, and a guidance scale of 3.5 is employed during sampling to enhance conditional generation.

B. Benchmark evaluation

The fidelity of the generated ECG-like signals is assessed using two quantitative metrics: Root Mean Squared Error (RMSE) and Fréchet Distance (FD), following the evaluation protocols established in RDDM. RMSE measures point-wise waveform deviation, whereas FD evaluates distributional similarity in a learned feature space. Because these metrics emphasize different aspects of generation quality, we interpret them jointly rather than treating either one as a complete measure of clinical validity. All comparisons are performed on 4-second signal segments for consistency with prior work.

Results: Table I shows a metric-dependent pattern. MAGIC achieves lower FD than RDDM in most dataset-step configurations, indicating stronger distribution-level similarity to real ECG. The advantage is especially clear on BIDMC and MIMIC-AFib, where MAGIC consistently reduces FD across all evaluated sampling steps. For example, on MIMIC-AFib with T=500, FD decreases from 6.424 for RDDM to 0.693 for MAGIC, suggesting that the multi-grained conditional pathway helps guide the generated distribution toward the target ECG domain. MAGIC obtains lower RMSE than RDDM on BIDMC for all step settings and on WESAD for some settings, but RDDM has lower RMSE on several DALIA settings and on small-step MIMIC-AFib settings. These results indicate that MAGIC's primary advantage in Table I lies in distributional fidelity rather than uniformly lower point-wise error. The discrepancy is plausible because FD rewards global morphology and feature-space realism, whereas RMSE penalizes sample-level phase or amplitude differences even when the generated waveform remains plausible.

Increasing the number of diffusion steps generally keeps MAGIC's FD low or further improves it, while RDDM shows less stable FD behavior in some datasets. Nevertheless, the benchmark supports that MAGIC is efficient and FD-favorable in most settings, but point-wise RMSE remains dataset-dependent.

TABLE I
COMPARISON OF DIFFERENT METHODS ON WESAD, DALIA, BIDMC AND MIMIC-AFIB DATASETS.

Diffusion timesteps	WESAD				DALIA			
	FD (↓)		RMSE (↓)		FD (↓)		RMSE (↓)	
	RDDM	MAGIC	RDDM	MAGIC	RDDM	MAGIC	RDDM	MAGIC
T=10	10.787	**7.451**	0.261	**0.259**	1.526	3.855	0.176	**0.171**
T=20	11.854	**5.002**	0.254	0.255	1.545	2.033	0.101	0.167
T=50	11.028	**3.313**	0.257	0.261	1.457	**0.832**	0.100	0.163
T=200	8.387	**3.635**	0.273	**0.264**	1.280	**0.514**	0.156	**0.151**
T=500	8.803	**2.883**	0.276	**0.270**	2.933	**0.517**	0.149	0.162

Diffusion timesteps	BIDMC				MIMIC-AFib			
	FD (↓)		RMSE (↓)		FD (↓)		RMSE (↓)	
	RDDM	MAGIC	RDDM	MAGIC	RDDM	MAGIC	RDDM	MAGIC
T=10	7.130	**5.134**	0.288	**0.237**	5.767	**3.609**	0.206	0.245
T=20	7.642	**3.247**	0.283	**0.226**	6.059	**1.854**	0.212	0.237
T=50	7.679	**2.317**	0.281	**0.226**	5.855	**0.958**	0.217	0.231
T=200	6.262	**2.034**	0.279	**0.251**	5.829	**0.716**	0.249	**0.229**
T=500	6.544	**2.022**	0.279	**0.251**	6.424	**0.693**	0.257	**0.229**

C. Utility evaluation

To evaluate whether the generated ECG-like signals preserve task-relevant information, we assess them across four downstream proxy tasks. These experiments provide indirect evidence about rhythm and morphology preservation under specific classifiers or regressors.

1) Heart rate estimation: To evaluate temporal information preservation in the generated ECG-like signals, we perform heart rate (HR) estimation on the DALIA and WESAD datasets. Following the established protocol used in CardioGAN [19], we apply the Hamilton algorithm [38]—a rule-based method that requires no training—directly to 8-second non-overlapping ECG-like windows. Accuracy is quantified by the Mean Absolute Error (MAE) in beats per minute between the HR derived from the generated and ground-truth ECG signals.

TABLE II
HEART RATE ESTIMATION PERFORMANCE USING GENERATED ECG-LIKE SIGNALS ON THE WESAD AND DALIA DATASETS.

Dataset	Test Modality	Method	MAE (↓)
WESAD	Orig. PPG	Reiss et al. [29]	11.305
	Gen. ECG	CardioGAN [19]	9.612
	Gen. ECG	RDDM [26]	3.351
	Gen. ECG	**MAGIC (Ours)**	**3.276**
DALIA	Orig. PPG	Song et al. [34]	11.453
	Gen. ECG	CardioGAN [19]	10.811
	Gen. ECG	RDDM [26]	5.426
	Gen. ECG	**MAGIC (Ours)**	**4.735**

Results: As shown in Table II, MAGIC achieves the lowest HR MAE on both datasets: 3.276 bpm on WESAD and 4.735 bpm on DALIA. Compared with RDDM, the relative reductions are modest on WESAD and larger on DALIA, suggesting that MAGIC can preserve R-peak timing information useful for HR estimation. Because HR is primarily determined by beat timing, this task supports the temporal fidelity of the generated signals, but it does not by itself validate full ECG morphology or diagnostic quality.

The improvement is consistent with the model design: token-wise PPG conditions provide local timing cues through gated cross-attention, while global conditioning helps stabilize rhythm-level context.

2) Atrial fibrillation detection: To evaluate whether generated ECG-like signals preserve AFib-related temporal and spectral information, we use AFib detection as a downstream proxy task. Following the evaluation protocol of RDDM, 4-second segments are transformed into spectrograms using short-time Fourier transform (STFT) and fed into a VGG-13 classifier pretrained on the MIMIC-AFib dataset. The classifier is trained for 25 epochs using the AdamW optimizer with a learning rate of 1×10^{-4} and a batch size of 64. Performance is measured by F1 score and accuracy, with real ECG results reported as a reference upper bound rather than as evidence of clinical equivalence.

TABLE III
AFIB DETECTION ACCURACY USING DIFFERENT MODALITIES AND METHODS ON THE MIMIC-AFIB DATASET.

Dataset	Test Modality	Classifier	Acc. (↑)	F1. (↑)
MIMIC-AFib	Orig. ECG	VGG-13	**0.706**	**0.693**
	Orig. PPG	Shen et al. [35]	0.424	0.432
	Gen. ECG (RDDM)	VGG-13	0.633	0.681
	Gen. ECG (MAGIC)	VGG-13	0.661	0.675

TABLE IV
PERFORMANCE COMPARISON ON THE PPG-BP DATASET FOR DIABETES DETECTION.

Dataset	Test Modality	Classifier	Acc. (↑)	F1. (↑)
PPG-BP	Orig. PPG	Avram et al. [36]	0.621	0.385
	Gen. ECG (RDDM)	VGG-11	0.764	0.453
	Gen. ECG (MAGIC)	VGG-11	**0.776**	**0.714**

Results: Table III shows that original ECG remains the strongest reference for AFib detection, with an accuracy of

0.706 and an F1 score of 0.693. Direct use of raw PPG performs substantially worse, which is expected because PPG does not directly contain ECG-specific features such as P waves and is more vulnerable to motion and perfusion artifacts.

Among methods utilizing generated ECG signals, MAGIC achieves an accuracy of 0.661, surpassing RDDM (0.633) by a relative improvement of nearly 3 percentage points. This performance advantage can be primarily attributed to the inherent sensitivity of AFib detection to irregularities in RR intervals. By incorporating a multi-scale feature extraction strategy into the diffusion process, MAGIC more effectively preserves fine-grained inter-beat rhythmic variations, thereby mitigating the over-smoothing artifacts that RDDM tends to exhibit when processing irregular rhythms. Consequently, MAGIC generates ECG signals that more closely approximate the true underlying AFib rhythm patterns.

3) Diabetes detection: To assess whether the generated ECG-like signals preserve information correlated with disease-related waveform changes, we evaluate diabetes detection as a downstream proxy task. Experiments are conducted on the PPG-BP dataset [32] using a VGG-11 classifier trained on spectrograms derived from 2-second ECG windows via short-time Fourier transform (STFT). The classifier is trained for 50 epochs using the AdamW optimizer with a learning rate of 1×10^{-4} and a batch size of 64. Performance is measured by F1 score and accuracy.

Results: As presented in Table IV, raw PPG yields an accuracy of 0.621 and an F1 score of 0.385. Both ECG-generation methods improve over this PPG baseline, and MAGIC obtains the highest values in this experiment, with an accuracy of 0.776 and an F1 score of 0.714. We interpret this result as evidence that MAGIC preserves features useful to the trained classifier on this dataset. However, because the evaluation is task-, dataset-, and classifier-dependent, it should be viewed as proxy evidence of information preservation rather than direct clinical validation.

4) Blood pressure estimation: To evaluate whether generated ECG-like signals preserve information correlated with cardiovascular dynamics, we perform blood pressure (BP) estimation as a downstream proxy task. Experiments are conducted on the Cuffless-BP dataset [33] using a 1D-UNet regressor trained on raw 10-second ECG and PPG signals. The model is trained for 50 epochs with the Adam optimizer (learning rate 5×10^{-4}) and a batch size of 512. Performance is quantified by Mean Absolute Error (MAE) for both systolic (SBP) and diastolic (DBP) pressure. SBP and DBP are affected by different physiological factors. This task assesses whether generated signals provide useful auxiliary input for the regressor, not whether they are clinically interchangeable with measured ECG.

Results: As presented in Table V, the benchmark approach combining original PPG and original ECG achieves notably low errors, with an SBP MAE of 3.670 mmHg and a DBP MAE of 0.763 mmHg. Among the methods incorporating generated ECG signals, RDDM achieves an SBP MAE of 2.952 mmHg and a DBP MAE of 1.550 mmHg. In comparison, MAGIC demonstrates superior performance in DBP estimation, achieving an MAE of 1.342 mmHg—a 13.4% reduction relative to RDDM. For SBP estimation, MAGIC attains an MAE of 3.143 mmHg, which is comparable to that of RDDM (2.952 mmHg), with a marginal difference of only 0.191 mmHg.

These findings highlight two key advantages of MAGIC. First, they exhibit a distinct strength in preserving morphological features essential for DBP estimation, including fine-grained ST–T segment morphology and QT interval dynamics. These features are intricately linked to peripheral vascular resistance and arterial compliance, both of which are critically involved in diastolic pressure regulation. Second, MAGIC achieves high fidelity in reconstructing systolic pressure-related features — such as QRS complex morphology and R-wave amplitude—with performance on par with RDDM, thereby enabling accurate assessment of cardiac contractile function.

D. Qualitative Visualization

To qualitatively assess waveform characteristics of the generated ECG-like signals, we performed a visual comparison between the best-performing RDDM configuration (T=50) and MAGIC. Representative samples were selected from four datasets: WESAD, DALIA, BIDMC, and MIMIC-AFib. These datasets cover a wide range of PPG acquisition conditions, including controlled laboratory settings, daily activities with motion artifacts, intensive care recordings, and pathological states such as atrial fibrillation. For each sample, we present a side-by-side visualization of the input PPG, the ground-truth ECG, and the corresponding outputs generated by RDDM and MAGIC. To further highlight the comparative performance, we selected representative cases from both AFib and Non-AFib groups as references.

Results: We qualitatively evaluate MAGIC and RDDM along three visual dimensions in Figure 3. **1) QRS morphology and R-wave sharpness:** MAGIC often produces sharper R waves and cleaner QRS complexes in the selected examples, whereas RDDM shows ringing or unstable amplitude in some cases. **2) ST–T morphology and baseline stability:** MAGIC tends to produce smoother baselines and more continuous post-QRS morphology in these examples, although visual inspection cannot establish diagnostic correctness. **3) Rhythm regularity and temporal coherence:** The selected MAGIC waveforms show RR timing that is visually closer to the reference ECG, especially in irregular rhythm examples. These observations support the quantitative FD results but remain qualitative and sample-dependent.

As shown in Figure 4, the AFib and Non-AFib examples illustrate that MAGIC can preserve some rhythm-related differences while reducing non-physiological high-frequency artifacts relative to RDDM in the displayed samples. We avoid interpreting these visualizations as diagnostic evidence; they are intended only to illustrate waveform characteristics.

E. Inference Time Comparison

We compared the inference time of RDDM and MAGIC under three different diffusion step configurations. The evaluation was conducted on the PPG test sets of four datasets by

TABLE V
PERFORMANCE COMPARISON ON THE CUFFLESS-BP DATASET FOR BLOOD PRESSURE ESTIMATION.

Dataset	Test Modality	Estimator	MAE	
			SBP (↓)	DBP (↓)
Cuffless-BP	Orig. PPG+Orig. ECG	UNet	3.670	0.763
	Orig. PPG	Vardhan et al. [37]	5.882	3.191
	Orig. PPG+Gen. ECG (RDDM)	UNet	**2.952**	1.550
	Orig. PPG+Gen. ECG (MAGIC)	UNet	3.143	**1.342**

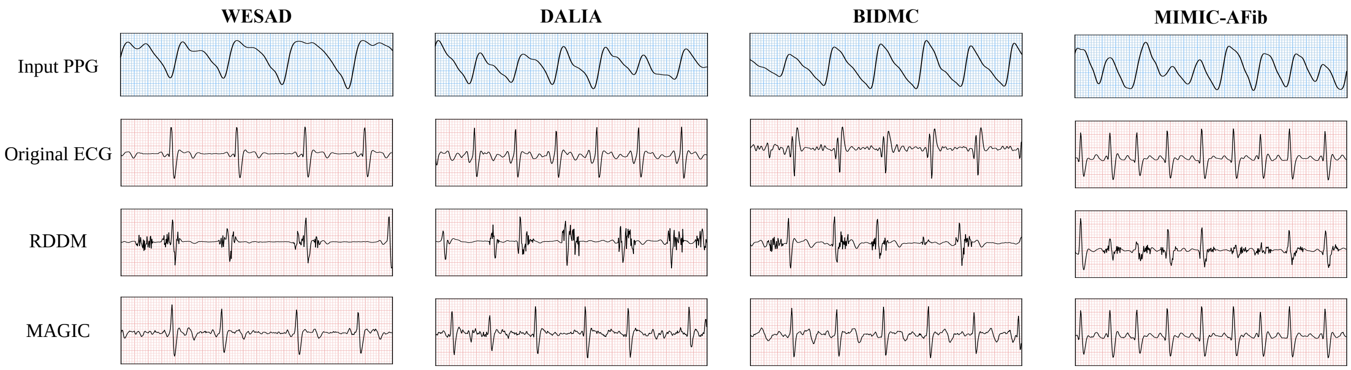


Fig. 3. Qualitative comparison of generated ECG-like signals on the WESAD, DALIA, BIDMC, and MIMIC-AFib datasets.

TABLE VI
COMPARISON OF DIFFERENT PREDICTION TYPES ACROSS FOUR DATASETS.

Prediction Type	WESAD		DALIA		BIDMC		MIMIC-AFib	
	FD (↓)	RMSE (↓)	FD (↓)	RMSE (↓)	FD (↓)	RMSE (↓)	FD (↓)	RMSE (↓)
v-prediction	**2.883**	**0.270**	0.517	0.162	2.022	**0.251**	0.693	**0.229**
x_0-prediction	2.956	0.319	2.144	0.186	3.163	0.277	1.560	0.254
ε-prediction	2.911	0.299	**0.485**	**0.161**	**1.991**	0.280	**0.685**	**0.229**

TABLE VII
ABLATION STUDY OF KEY COMPONENTS IN MAGIC ACROSS FOUR DATASETS.

Module	WESAD		DALIA		BIDMC		MIMIC-AFib	
	FD (↓)	RMSE (↓)	FD (↓)	RMSE (↓)	FD (↓)	RMSE (↓)	FD (↓)	RMSE (↓)
Pretrain	**2.883**	**0.270**	**0.517**	**0.162**	**2.022**	**0.251**	**0.693**	0.229
w/o pretrain	2.325	0.309	0.598	0.268	2.251	0.292	0.908	**0.223**
w/o tok	2.988	0.301	0.600	0.264	2.085	0.279	0.731	0.230
w/o glb	3.169	0.303	0.712	0.265	2.099	0.279	0.759	0.230

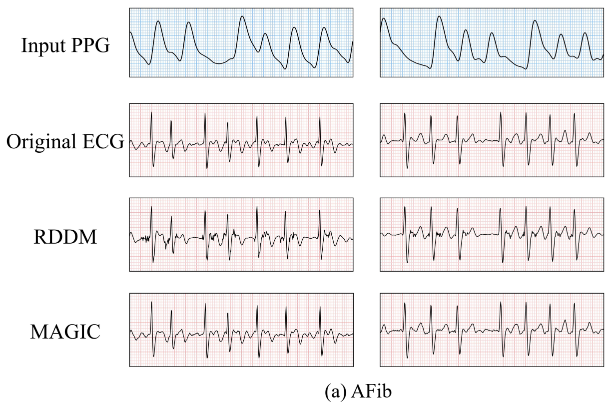
measuring the total time required to convert all test samples into ECG-like signals, corresponding to the inference latency used in the standard FD evaluation. All experiments were performed using a single GeForce RTX 4090 GPU.

Results: Figure 5 compares inference efficiency and FD under different sampling-step settings. At the same step count, MAGIC requires less inference time than RDDM in our implementation. Moreover, MAGIC at T=50 obtains lower FD than RDDM at T=500, indicating favorable sampling efficiency. This result should be interpreted together with Table I: the efficiency advantage is most clearly reflected in FD, whereas point-wise RMSE remains dataset-dependent.

F. Prediction Type Analysis

In the design of diffusion models, the choice of prediction target is a pivotal factor influencing both generation quality and training stability. Common formulations include direct prediction of the original data (x_0-prediction), prediction of the added noise (ε-prediction), and velocity prediction (v-prediction). Each prediction target corresponds to distinct weighting strategies and sampling dynamics, which can substantially affect the model's ability to capture high-frequency details and global signal structure. To investigate this, we conduct comparative experiments under the proposed MAGIC framework, evaluating the three prediction types across four datasets—WESAD, DALIA, BIDMC, and MIMIC-AFib—using FD and RMSE as quantitative metrics, to examine the trade-offs of the prediction strategy adopted in MAGIC.

Results: Table VI shows that v-prediction provides a strong overall trade-off, but it is not optimal for every individual metric. For example, ε-prediction obtains slightly lower FD on DALIA, BIDMC, and MIMIC-AFib, and a slightly lower



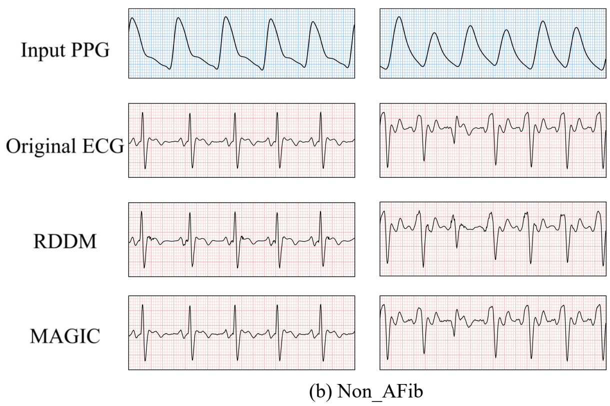


Fig. 4. Side-by-side visualization of generated ECG-like signals from AFib and Non-AFib groups.

RMSE on DALIA; the RMSE on MIMIC-AFib is tied. In contrast, v-prediction improves over x_0-prediction on all reported FD/RMSE entries and remains close to the best value in most datasets. We therefore use v-prediction as a balanced and stable choice rather than claiming that it is universally optimal across all metrics.

G. Ablation Studies

By comparing the full model (Pretrain) with variants that remove pre-training (w/o pretrain), token-wise conditioning (w/o tok), and global conditioning (w/o glb) across four datasets, we quantify how each component affects FD and RMSE.

Results: Table VII shows that the contribution of conditional-encoder pre-training is dataset-dependent. Pre-training improves FD on DALIA (0.517 vs. 0.598), BIDMC (2.022 vs. 2.251), and MIMIC-AFib (0.693 vs. 0.908), indicating that cross-modal alignment is helpful in several settings, especially the pathological MIMIC-AFib dataset. However, the variant without pre-training achieves a lower FD on WESAD (2.325 vs. 2.883) and a slightly lower RMSE on MIMIC-AFib (0.223 vs. 0.229).

The ablations also support the importance of both conditional granularities. Removing token-wise conditions worsens most FD/RMSE entries, while removing global conditioning leads to clear FD degradation, especially on DALIA and MIMIC-AFib. These results are consistent with the intended design: token-wise conditions provide local temporal cues, and global conditioning stabilizes segment-level denoising dynamics.

V. Discussion and Limitations

Several limitations should be noted. First, PPG-to-ECG translation is inherently ill-posed: different ECG morphologies may correspond to similar peripheral pulse waveforms. The generated output should therefore be regarded as an ECG-like auxiliary representation rather than a substitute for clinically acquired diagnostic ECG. Second, downstream tasks such as HR estimation, AFib detection, diabetes detection, and BP estimation are proxy evaluations of information preservation under specific models and datasets; they cannot establish clinical equivalence or diagnostic safety. Third, the ROI mask is derived from ground-truth ECG and is used only during conditional-encoder pre-training, which may limit its applicability when paired ECG is unavailable. Finally, the present study does not include statistical significance testing, multi-seed uncertainty analysis, prospective validation, or expert cardiologist assessment. Future work will investigate uncertainty-aware generation, morphology-specific validation, subject-independent and cross-device generalization, and prospective clinical studies.

VI. Conclusion

This paper presents MAGIC, a multi-grained conditional diffusion model for PPG-guided ECG-like waveform generation. The method aligns PPG and ECG representations through conditional-encoder pre-training, separates global and token-wise conditions, injects global context through AdaLN, and injects local PPG information through gated cross-attention. Experiments across four public datasets show that MAGIC improves FD in most benchmark settings and offers favorable sampling efficiency. Downstream proxy tasks further suggest that the generated signals can preserve useful rhythm- and morphology-related information, although the outcomes are mixed for AFib F1, SBP estimation, RMSE, and some ablation metrics. These results support MAGIC as an auxiliary computational tool for PPG-based waveform generation, while clinical diagnostic use requires further validation with real ECG references and prospective studies.

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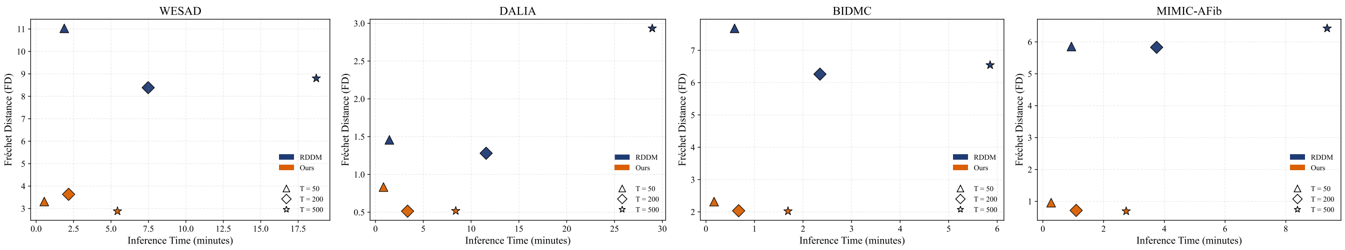

Fig. 5. Inference time and FD comparison between RDDM and MAGIC under different sampling-step settings. Lower FD indicates better distributional similarity, and shorter time indicates higher sampling efficiency.

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